

around principles formulated by people like Weber, informed by logic and the model of the machine. Network-pathologies will arise in network organizations which are built around principles formulated by people like Nolan and Crosan, informed by logic and the model of the IT system.

All that said, I think it is fair to conclude the following. Nolan and Croson provide a useful framework which, at worst, can be used as a coherent, if somewhat general and idealistic, starting point for theoretical development and as a source of ideas for practical management. At best, it has a good lock on what the organization of the future will be like and therefore can be of immense value to both managers and academics. In any event, the book is well written, easy and interesting to read, and a provocative discussion of many of the issues that face organizations in the nineties.

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National innovation systems: a comparative analysis, Edited by Richard R. Nelson, Oxford University Press, New York, 1993. ISBN 0 19 507616 8 Hdbk, 019 507617 6 Pbk, Price: £40.00 Hdbk, £25.00 Pbk.

It has taken me a long time to read all of this large book! This is not because I am a slow reader or a lazy reviewer (though the reviews editor may take a different line on this!) but because it is not necessary. Indeed one of the strengths of this book is that you *should not* want or need to read it all the way through at one go. It is to be dipped into as a series of fascinating (if sometimes over-detailed) essays on how 15 countries have organised, currently organise and might have to re-organise support for the development, commercialisation and diffusion of new products, processes and services; in short those nations' 'systems' for the promotion of technological innovation.

It is the outcome of an international project, calling upon a panel of prestigious

analysts of technology policy to give an account of their own country's innovation and research system. Covering 15 countries means that the book does not confine itself to the 'usual suspects' dealt with by other books on technology policy in English: namely the USA and/or Britain in comparison with Japan. There are indeed well-written chapters on these countries but also on other 'market-oriented' industrialised, high-income countries, both large (Germany, France and Italy) and small (Denmark, Sweden, Canada and Australia). Even more unusually, there are detailed accounts of the national innovation systems of newly-industrialising countries of East Asia — South Korea and Taiwan — and of Brazil, Argentina and Israel.

The accounts follow roughly the same structure: an historical section on the industrial and technological development of the country, an outline of the present system (covering trade and industrial output, state education/training and technology support policies) and a brief conclusion of the challenges the country faces in coping with increasing internationalisation of trade and production and assorted political problems. Each chapter has a veritable mine of references, invaluable for researchers of international technology policy, especially of the least-examined countries. One chapter, on Taiwan, was especially useful to your reviewer, in briefing him on that country's technological strengths and policies for a recent visit. In addition, the chapters have proved very useful for university student seminars: the *range* of countries discussed counteracts the impression often gained (at least in the UK) that the only system that has anything to teach us about the best way to organise for innovation support is Japan's. There is much to be learned from Germany in organising industrial innovation at a regional level, from France in making state-controlled industries successfully innovative and from Italy in networking innovation amongst small firms.

The chapters concentrate on the *direct* factors that constitute an innovation system: education/training; government subsidy of industrial R&D, innovation and diffusion; support for basic research and technology transfer; technological procurement (in

defence, space, energy, health care, etc.). They only tangentially refer to important influences from more general state policies, such as policies on taxation or financial management and they avoid completely taking refuge in that scoundrel of (non-) explanations of country differences: namely 'culture'. There is however a big omission in the discussion of the direct factors that strongly influence a nation's innovation, namely *regulation*. The particular regimes for preserving the environment, health and safety etc. that a country has in place can be very influential in affecting the rate and direction of innovation, and despite increasing global or regional harmonisation of these regulations there are still strong national differences in the regulations themselves and, more crucially, in enforcement. Perhaps another book is needed to deal with this topic.

Though readers are expected to make their own comparisons between the countries, they get some excellent guidance in a concluding chapter by Richard Nelson. He admits that the rapidly growing internationalisation of trade, production and even new-technology development could be seen as dissolving the 'National' basis of Innovation Systems. Internationalisation of technology-based industrial processes/products means that 'the competitive edge will depend on the details of design, production process, firm strategy and organisation, upstream-downstream connections...'. So, some consider that the attempts of national governments to define and support national high-technology industry will be frustrated because of this internationalisation. But, as Nelson points out, being 'constrained' by factors such as the activity of transnational companies, outside your national boundaries, is hardly novel; since 1945, all countries, and especially small high-income and 'third world' countries, have always been obliged to take account of international trading rules and global political realities. Despite the internationalisation trends, much of what influences how new products/processes/services are innovated and diffused remains nationally-peculiar; this is especially true of scientific and technical education and training, the organisation and funding of basic research and mechanisms of technology

transfer: these remain national almost by definition. Then there are a whole range of other national 'associated systems' — for example, the law and taxation — with a strong bearing on innovation systems. Internationalisation or Globalisation may require that national systems *change*, but they will still be specifically *national*. As such, countries have a lot to learn from each other. This book, read slowly, will help.

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Simplicity wins. How Germany's mid-sized industrial companies succeed. G. Rommel et al., Harvard Business School Press, Boston, MA, 1995, ISBN 087-584-504-5. \$27.95 hardcover.

Company success can be achieved by avoiding complexity and by choosing 'simple goals, structures, and procedures, selected and designed without compromise, and rigorously implemented' (pp. 2, 10). This message is derived from an analysis of the performance of 39 medium sized companies in the German mechanical engineering and component manufacturing industries during the late 1980's as well as illustrative case material. Success is a construct made up from return on sales, sales growth, and cash flow over sales (pp. 4/5). While the book covers chapters on vertical integration, logistics, production technology and organization, its largest part deals with development. Major messages in this chapter are to avoid overengineering, to invest in predevelopment phases, to implement smallstep innovations continuously, and to establish an appropriate organization structure, particularly to avoid classical functional development organizations.

The book is translated from an earlier German publication, and a short chapter is added (8 pages) on a comparison with 21 component manufacturers in the US.

The book is written for practitioners, and it integrates most of the management wisdom that is available today in a compressed way. It does not draw on the impressive amount of literature that is available on